Velocity

LEARNING TARGET · S1 I can use a position–time graph's slope to find velocity.

SECTION A KEY VOCABULARY

Velocity	*How fast and which way.* The rate position changes.
Speed	*How fast only.* The size of the velocity, with no direction.
Slope	Steepness of a line: rise ÷ run. On a position–time graph, slope = velocity.
Average speed	Total distance ÷ time. A scalar — no direction. Velocity adds the sign.

SECTION B KEY EQUATIONS + UNITS

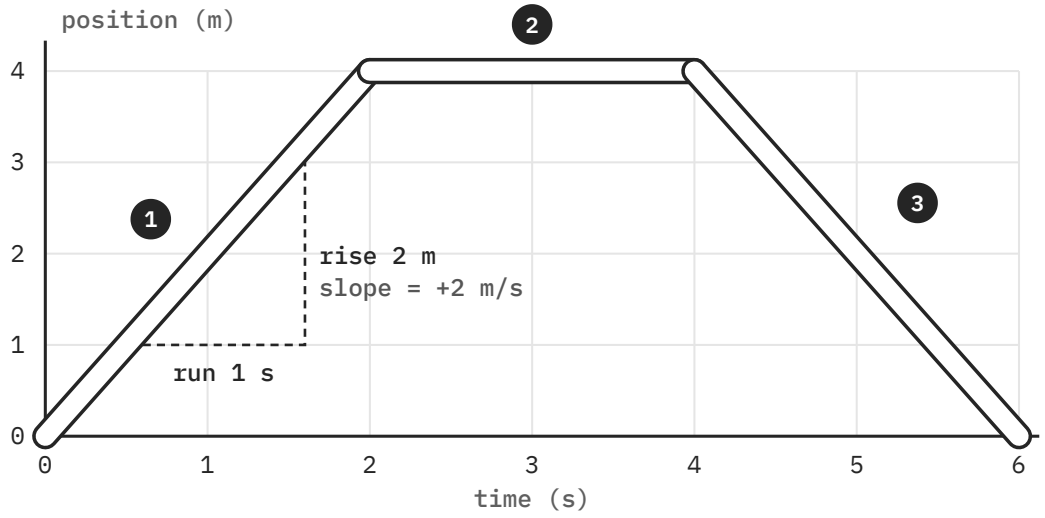
$$v = \Delta x \div \Delta t$$

speed = distance ÷ time

v	m/s (vector: has a sign)
convert	1 m/s = 3.6 km/h

SECTION C COLOR IT · READ A GRAPH AS A MOTION STORY

Height = position. Slope = velocity.



☐ **1 · PURPLE** Rising: moving +, v > 0

☐ **2 · PURPLE** Flat: stopped, v = 0

☐ **3 · PURPLE** Falling: moving –, v < 0

SECTION D WORKED EXAMPLE

A cart goes from 8 m to 2 m in 3 s. Right is positive. Find its velocity.

GIVEN	EQUATION	WORK	ANSWER
x_i = 8 m x_f = 2 m Δt = 3 s	v = Δx ÷ Δt	Δx = 2 − 8 = −6 m v = −6 m ÷ 3 s	**−2 m/s** 2 m/s to the left

SECTION E THE 2026-XJ THREAD

DAY 3 OF 22 · NEW TOOL: VELOCITY

Two position readings and the time between them give XJ's velocity: the slope of its position–time graph. That tells us how fast it closes in, and from which side.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else